

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8509**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600061
Course Section:	1				

L_29_518_986_L_1505_38449



Name of the candidate. Shubham Rajapurkar

Candidate's UID number:

2	0	2	2	6	0	0	0	6	1
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Examination class room number:

4	0	1
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Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE - BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: CSE - AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: WEDNESDAY 08/10/25

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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(Begin answer for each question on a new page)

Q1a]

$$f(w) = \frac{1}{2} \|xw - y\|^2$$

$$o(w) = \bar{w} \bar{x}$$

$$E(w) = \frac{1}{2} \sum (t_d - o_d)^2$$

$$E = \left(\frac{\partial E}{\partial w_0} + \frac{\partial E}{\partial w_1} + \dots + \frac{\partial E}{\partial w_n} \right) \quad \text{learning rate}$$

$$w_i = w_i + \Delta w$$

$$\Delta w = -\eta \frac{\partial E}{\partial w}$$

change in weight

$$\frac{\partial E}{\partial w} = \frac{1}{2} \sum (t_d - o_d)^2$$

$$= \frac{1}{2} \sum (t_d - o_d)^2$$

$$= \frac{1}{2} (x(t_d - o_d) \sum (t_d - o_d))$$

$$= (t_d - o_d) \sum (t_d - w \cdot x_d)$$

$$\frac{\partial E}{\partial w} = (t_d - o_d)(-x_d)$$

$$\Delta w = -\eta \frac{\partial E}{\partial w}$$

$$= \eta (t_d - o_d)(x_d)$$

$$\Delta w = \eta (t_d - o_d)(x_d)$$

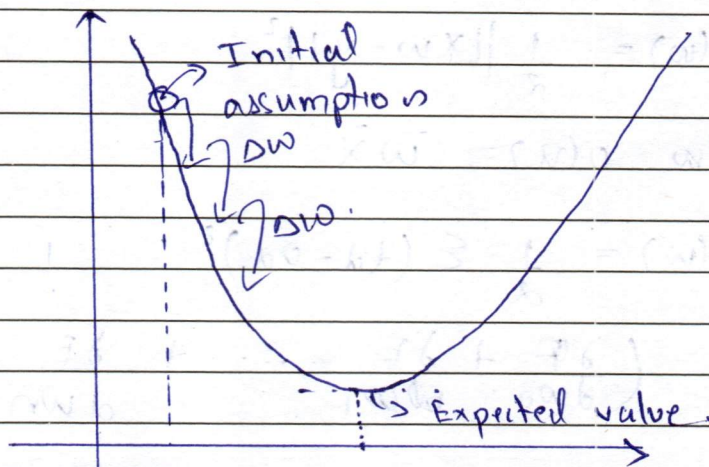


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The convergence is required till we reach a minima and the error in the gradient descent is ~~low~~ the lowest.



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	Q1.b	The bias - variance decomposition states that if the bias is high variance will be low and if the variance is high bias will be low. The expected error for this can be given by - $E = \frac{1}{2} (n - n_i)^2$ When the model capacity is too high it may lead to overfitting as the model becomes too generalized and work great on the training set and fail to perform on the testing set. It may happen when the no. of parameters are too high and the mean square error between the is too low. As a result the model fails to classify it between the output. Such a model have high a high bias and low variance.

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Q2a

The input size $\Rightarrow H \times w \times C$ filter size $\Rightarrow F \times F$ no. of filters $= K$

Dimension of resultant filter

$$\hookrightarrow (H-F+1)(w-F+1) \times C$$

In a standard 2D convolution the computational complexity is very high as first we need to calculate the output for each pixel and then we have to do that for the entire height of the input image.

as a result the no. of multiplications are quite high.

$$\hookrightarrow (H-F+1)(w-F+1) \times C$$

A depth separable convolution solves the ~~for~~ problem as it reduces the image dimension ~~significantly~~ significantly thus ~~or~~ thus reducing the computational complexity and the no. of multiplications.

Eg. if the filter is of size $F \times F \times F$

The dimension of output

$$\hookrightarrow (H-F+1)(w-F+1)(C-F+1)$$

Thus Hence there is a significant reduction in complexity



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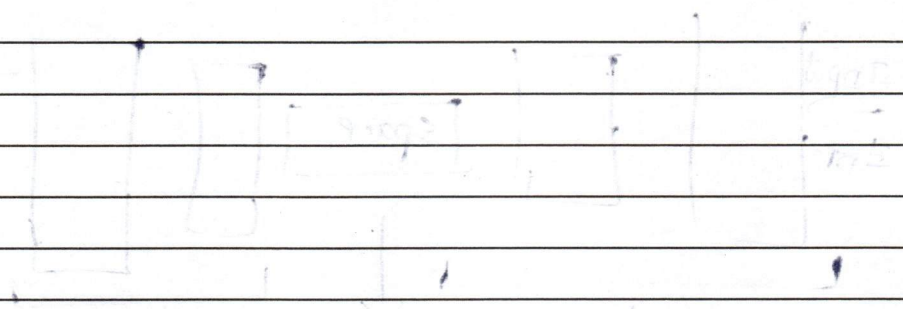
Question
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Because of the benefits of faster calculation less complexity and less no. of multiplications

It forms a core of modern architectures

The depthwise separable convolution has a far less parameters efficiency than improve the speed and performance.



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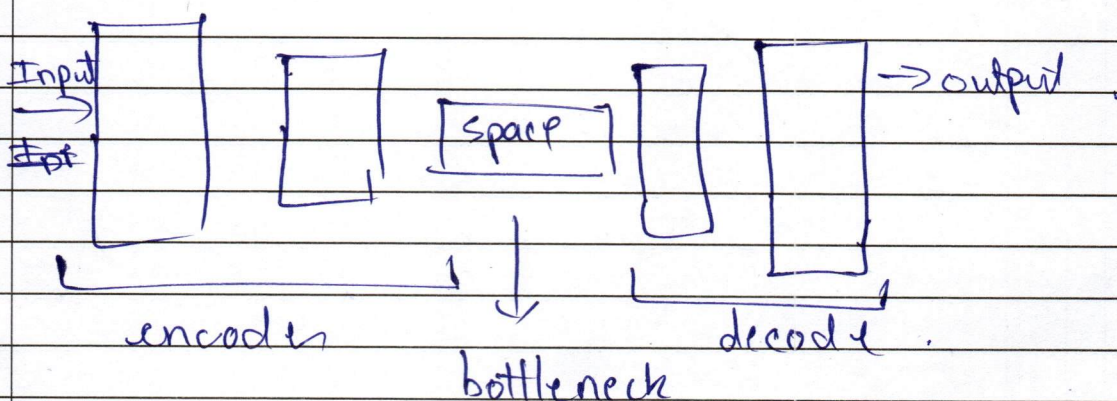
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Q26

The In the linear undercomplete autoencoder
the ~~dim~~ $\dim(\text{space}) < \dim(\text{input})$

As a result the undercomplete autoencoder help to reduce feature from the input. and is use for dimensionality reduction.

The PCA performs the same task of dimensionality reduction, by reducing the no. of feature from the input.



~~dim bott~~ $\boxed{\dim(\text{space}) < \dim(\text{input})}$

In a overcomplete auto encoder, the
 $\dim(\text{space}) > \dim(\text{input})$

It This lead expansion of feature as the $\dim(\text{space})$ is high

It may lead to overfitting of the model without regularization

Thus, a proper regularization is required. so that the model work proper and the overfitting is kept in check.



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	Q3a	In a LSTM, It uses gates to modify the var information thus can control what unfor should be kept in the memory and what should be forget The LSTM consist of 3 gates. forget gate Input gate. Output gate. and 2 memory. long term memory and short term memory. The forget gate helps to us to forget information from the long term memory. $f_{t+1} = f(w_t[h_t, u_t] + b_f)$ ↓ 0 $[f_t \otimes C_t]$ Single Since there are 2 memory long term and short term and gates approach It help in prevent the vanishing gradient problem.

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Q36 The GRU consist of 2 gates, update gate, reset gate.

The update gate is use to forget the value from the memory.

and reset gate is use to set the values in the memory.

The GRU has a single hidden state as compared the 2 state in the LSTM, it consists of a hidden state < > ~~cell~~ and current state.

As a result the GRU consist of far less parameter than the LSTM ~~cell~~ while the performance of both stay quite comparable.

Since there are far less parameters in the GRU. ~~as~~ The GRU has a greater real word applications as it is less complex than the LSTM and equally in performance.

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